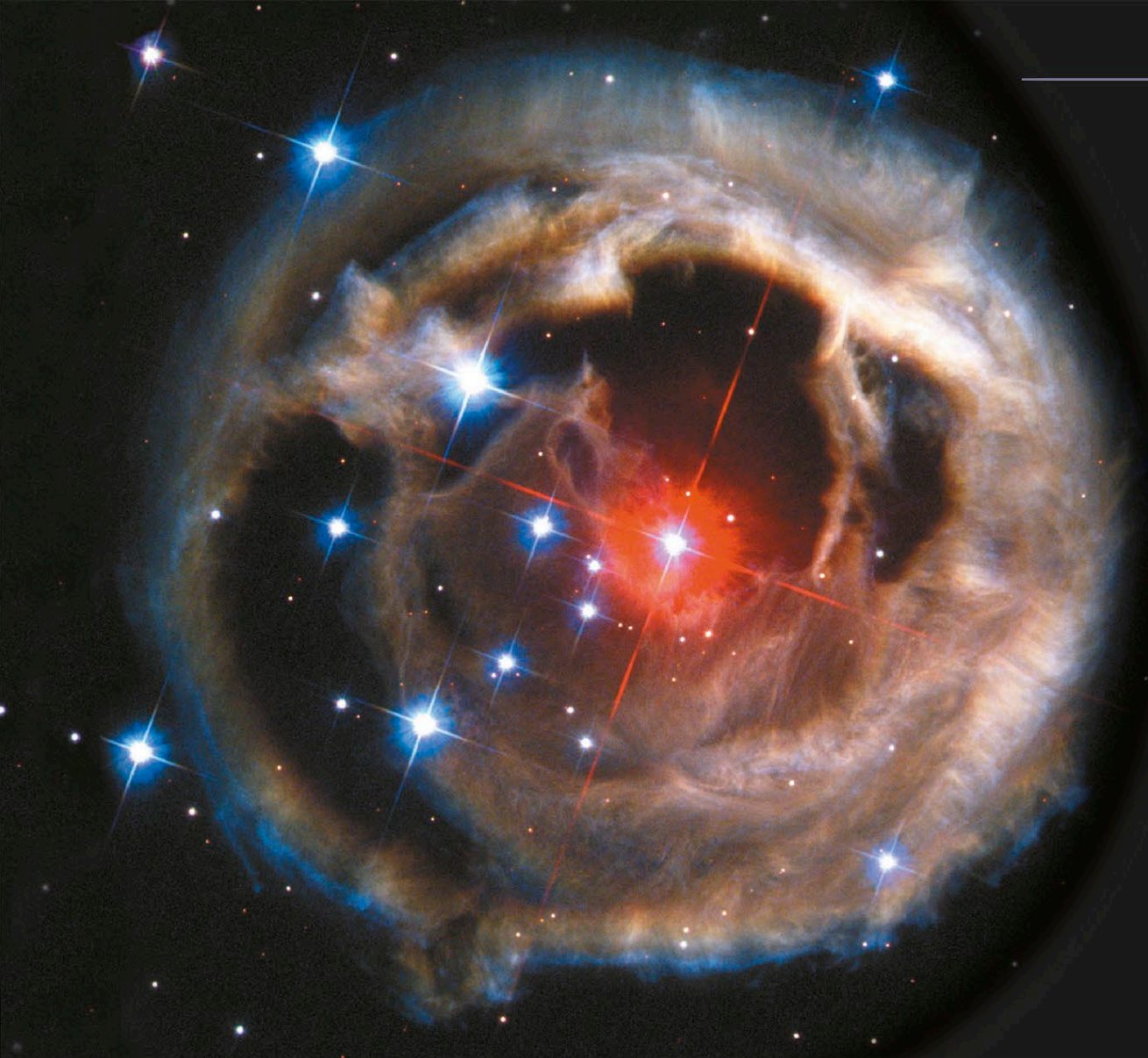




RED SUPERGIANT

V838 Monocerotis



DISTANCE FROM SUN
20,000 light-years
MAGNITUDE 10
SPECTRAL TYPE K

MONOCEROS

Discovered on January 6, 2002, by an amateur astronomer, V838 Monocerotis is one of the most interesting stars. Its precise nature is not yet fully understood, but astronomers believe its recent evolution has moved it off the main sequence to become a red supergiant. While this phase would usually take hundreds or thousands of years, here it has happened in a matter of months. Its first viewed outburst, in January 2002, was followed a month later by a second in which it brightened from magnitude 15.6 to 6.7 in a single day—an increase of several thousand times. Finally, in March 2002, it brightened from magnitude 9 to 7.5 over just a few days. The energy emitted in the outbursts caused previously ejected shells of material to brighten and become visible.

LIGHT ECHOES

Light echoes from recent outbursts illuminate the ghostly shells of ejected material around the enigmatic star V838 Monocerotis (seen glowing red).

BLUE SUPERGIANT

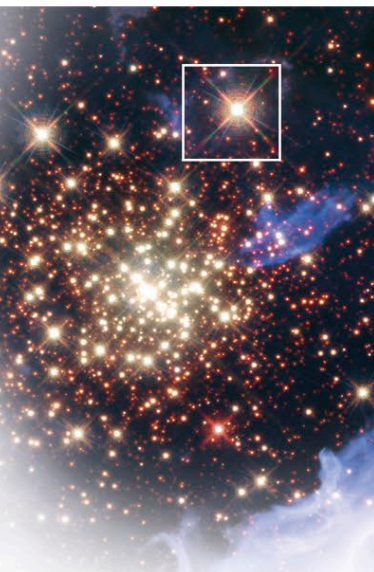
Sher 25



DISTANCE FROM SUN
20,000 light-years
MAGNITUDE 12.2
SPECTRAL TYPE B1.5

CARINA

This blue supergiant is poised to explode as a supernova, possibly within the next few thousand years. The prediction of its apparent closeness to death has been based on observations that reveal striking similarities between Sher 25 and Sk-69 202, the progenitor star of the supernova that occurred in the Large Magellanic Cloud in 1987 (now known as SN 1987A, see p.310). Sher 25 lies at the center of a clumpy ring of ejected material, and additional material from the star is escaping perpendicular to this ring. This has caused the ejected stellar material to form an hourglass-shaped nebula with Sher 25 lying at its middle. The ring and nebula are similar to those observed around Sk-69 202 before that blue supergiant exploded. Spectroscopy reveals that the nebula



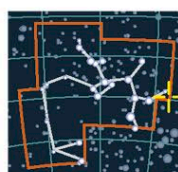
PENDING SUPERNOVA

The blue supergiant shown boxed in this image is likely to explode as a supernova. The open cluster of bright white stars and the surrounding nebula are known as NGC 3603.

surrounding Sher 25 is rich in nitrogen, indicating that it has passed through a red supergiant phase, again displaying an evolutionary path similar to that of the star Sk-69 202.

BLUE VARIABLE

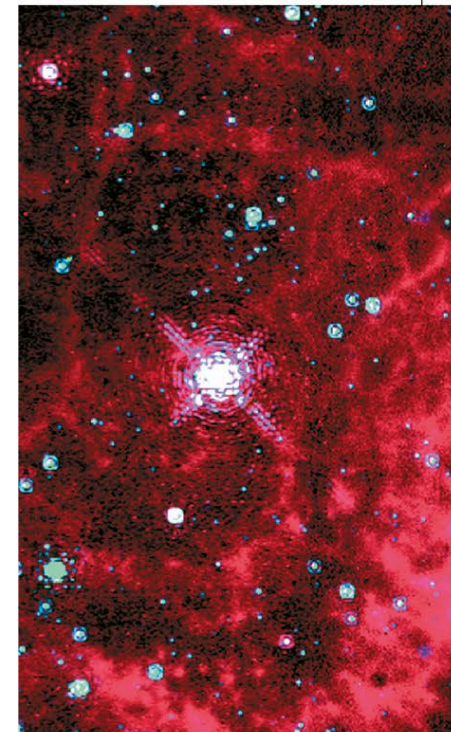
Pistol Star



DISTANCE FROM SUN
25,000 light-years
SPECTRAL TYPE LBV

SAGITTARIUS

One of the most luminous stars ever discovered is located at the center of the Pistol Nebula and is known as a luminous blue variable. The Pistol Star emits around 10 million times more light than the Sun, unleashing as much energy in 6 seconds as the Sun does in 1 year. When it first formed, it contained at least 100 solar masses of material, but a stellar wind billions of times more powerful than the Sun's, along with huge eruptions, have seen it shed at least 10 times the Sun's mass. The biggest eruptions occurred about 4,000 and 6,000 years before its presently seen state. In the Sun's position, the star would fill the diameter of Earth's orbit. Despite its size and luminosity, the star is obscured at visible wavelengths by the ejected material that has formed the pistol-shaped nebula surrounding it.



VAST NEBULA

Seen in infrared light, the Pistol Nebula glows a brightly. The nebula is 4 light-years across and would nearly span the distance from the Sun to Proxima Centauri, the closest star to the solar system.